

Diploma Supplement

Preamble

The purpose of the Diploma Supplement is to provide sufficient independent data to improve the international 'transparency' and fair academic and professional recognition of qualifications (diplomas, degrees, certificates etc.). It is designed to provide a description of the nature, level, context, content and status of the studies that were pursued and successfully completed by the individual named on the original qualification to which this supplement is appended. It is free from any value judgements, equivalence statements or suggestions about recognition. This Diploma Supplement model was developed by the European Commission, Council of Europe and UNESCO.

1. Information identifying the holder of the qualification

1.1 Last name(s)

SISSA

1.2 First name(s)

MATTEO

1.3 Date of birth(dd/mm/yyyy)

29/04/2001

1.4 Student identification number or code

152814

2. Information identifying the qualification

2.1 Name of the qualification and (if applicable) title conferred (in the original language)

Laurea in Computer Engineering (D.M.270/04)
Bachelor

2.2 Main field(s) of study for the qualification

Information technology engineering - L-8
ISCED CODE: 0714

2.3 Name (in original language) and status of the awarding institution

MODENA and REGGIO EMILIA University
Statal
Head Office: Via Università', 4 MODENA

2.4 Name and status of institution (if different from 2.3) administering studies (in original language)

Not applicable

2.5 Language(s) of instruction/examination

Mod. C_DS_C; Matr: 152814

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3. Information on the level and duration of the qualification

3.1 Level of qualification

1 ° cycle QF-EHEA /6° Level EQF

3.2 Official duration of the programme in credits and/or years

180 CFU/ECTS - Three full time years

3.3 Access requirement(s)

Italian secondary school leaving qualification or other comparable foreign qualification (level 4 EQF)
[TITLE OF HIGH SCHOOL]

Italian secondary school leaving qualification or other comparable foreign qualification (level 4 EQF)
[TITLE OF HIGH SCHOOL]

4. Information on the contents and results gained

4.1 Mode of study:

Full time - PRESENCE

Interactive lessons, Laboratory, Training debt, Lesson, Exercise

4.2 Programme learning outcomes

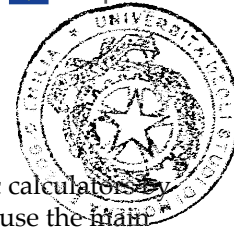
The Bachelor's Degree Programme in Computer Engineering is mainly aimed at providing graduates with an adequate mastery of general scientific methods and contents, which are useful to complete their own professional development profitably through subsequent training programmes and - when entering the job market - to adapt to the fast technological development that is typical of Computer Engineering and the production sectors it applies to. In order to achieve such priority training objective, the Bachelor's Degree Programme in Computer Engineering is aimed at providing graduates with a suitable training in the following learning fields: 1) Basic sciences for a sound preparation in mathematics, physics, and information technology, which are essential to interpret, describe and solve the issues of engineering in general and ICT engineering in particular; 2) IT for the Information and Communication Technology, to provide the knowledge of the basic principles of data processing systems and the skills essential to use them; 3) Other ICT subjects to offer a wide scope training in information engineering, aimed to provide the essential knowledge and skills in some of the additional subjects that are distinctive for the Degree Class, including electronics, telecommunications, and automation; 4) Similar and integrating subjects, for a suitable training in some similar and integrating subjects that are useful to provide further knowledge of a scientific and engineering nature. Based on the learning areas listed above, the Degree Programme in Computer Engineering includes a sound cultural and methodological training programme that can be integrated with customised study paths aimed at providing a type of training focusing on both accessing the job market and undertaking further studies for a Master's Degree, or a 1st level university master programme. To that purpose, the Degree Programme in Computer Engineering also offers its students the opportunity to: a) carry out training activities aimed at facilitating the professional choices



through the direct knowledge of the employment sector that students can access with this qualification, by participating in training internships (3 months); b) gain further skills in specific engineering fields; c) gain skills in subjects that are useful to understand the legal, social and ethical contexts of the engineering profession. As outlined in A2.a, the professional profile resulting from the 4 learning areas described above is of an IT Engineer focused on the design and production of software applications, and the processing of digital information. In accordance with the university qualification descriptor system adopted by the European Union (Dublin Descriptors), with the educational qualifying objectives of the Class of Degrees in Information Engineering, as well as the above-mentioned specific training objectives, the Degree Programme in Computer Engineering awards the final qualification to those students who have attained the learning results expected in the 4 areas mentioned above and further described below, together with the list of the training activities included in the master's degree programme.

Knowledge and understanding. 1) Basic sciences - Know and understand the main concepts of mathematical analysis relating to functions and differential and integer calculation, besides differential equations. - Know and understand the main concepts of linear algebra and two- and three-dimensional Euclidean geometry. - Know and understand the main concepts of physics, and electromagnetism in particular. - Know and understand the fundamentals of information technology, and in particular of the imperative and sequential programming. - Know and understand the main concepts of operating research relating to mathematical modelling for decision-making issues and the optimisation through programming in line with continuous, whole or mixed variables. 2) Information Technology for ICT - Know and understand the basics of electronic calculators, in particular those based on Intel microprocessors. - Know and understand the basics of operating systems, with specific reference to UNIX/Linux and related system programming. - Know and understand the basics of data bases, relational ones in particular. - Know and understand the basics of object-based programming, and Java in particular. - Know and understand the basics of software engineering. - Know and understand the basics of calculator networks and main application services. 3) Other ICT subjects - Know and understand the basics of electric circuits and digital electronics. - Know and understand the basics of telecommunications. - Know and understand the basic concepts of automated controls. 4) Similar and integrating engineering subjects - Know and understand the basics of statistics and mathematic probability. - Know and understand the thermal implications resulting from the operation of components, circuits, and electronic systems. - Know and understand company operations both from a management perspective and in terms of accounting reports of their results.

Applying knowledge and understanding. 1) Basic sciences - Ability to model and resolve mathematical problems using the mathematical analysis techniques, more specifically the functions of a variable, maximum and minimum points of functions with multiple real and integer variables of volume and surface. - Ability to model and resolve algebraic and geometrical problems using the techniques of linear algebra and geometry. - Ability to apply the basics of classic electromagnetism up to Maxwell equations. - Know how to define sequential algorithms and be able to develop and test programmes by using C programming language. - Ability to represent decision-making and optimisation issues through linear mathematic models, and to solve simple issues of continuous, whole, or mixed linear mathematic programming. It should be observed that the following list includes two training activities ending in "-IEI" that, even if they may slightly differ from the corresponding training activities with no suffix, they share the same contents and therefore fall within the same learning areas. Such training activities are carried out as service at the Military Academy of Modena, as a result of a specific convention between the University of Modena and Reggio Emilia and the Ministry of Defence. They are therefore reserved exclusively to students of the Italian Army. 2)

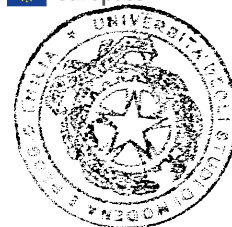


Information Technology for ICT - Ability to analyse the main architectures of electronic calculators by assessing their performance and to write simply functions in assembly. - Know how to use the main shell controls of UNIX/Linux and be able to develop script programmes in Shell and competing programmes by using C programming language. - Ability to design, create, change and query a relational database. - Ability to develop programmes by using the Java programming language. - Ability to apply the software engineering techniques to software systems specifications, analysis, and design. - Know how to use the main application services of computer networks and be able to install and set up local networks and Web servers in UNIX/Linux environment. 3) Other ICT subjects - Be able to analyse linear electric circuits and understand the behaviour of electronic circuit components within calculation systems. - Be able to design analogue and numerical links for telecommunications. - Be able to describe and analyse back-driven dynamic systems and design regulators aimed at improving the dynamic performance of back-driven systems. 4) Similar and integrating engineering subjects - Ability to apply statistics and mathematic probability. - Ability to apply balance integral equations and the dimensioning techniques of equipment and devices for heat dissipation and thermal insulation. - Know how to use the tools required for the economic and financial analysis and for interpreting management results.

Making judgements. The future IT Engineers are required to possess: a) The ability to collect and interpret data, and to formulate personal opinions on such data; b) The ability to understand the impact of engineering solutions on the social and physical, and environmental context. The Degree Programme in Computer Engineering is aimed at providing students with the suitable methodological and operational tools useful to independently and objectively deal with both the typical issues relating to the design and production of software applications, and the processing of digital data. The assessment of the expected results is carried out both in the single teachings and more specifically in those of the sector of the systems for the information processing (ING-INF/05), all providing for laboratory activities, both in the final examination and possibly in taking up an internship or a project activity.

Communication skills. The communication abilities required to a future IT Engineer specifically refer to the ability to communicate information, ideas, issues, and solutions to specialist and non-specialist counterparties, both orally and in writing, in Italian and in English. Such abilities (in Italian) are ascertained both through written and/or oral examinations provided in the single teachings, the drawing up and presentation of a final paper and, possibly, by taking up an internship or project activity. As regards communication skills in English in particular, besides possessing level B1 of the Common European Reference Framework, students may gain a further level B2 knowledge (by means of the teaching activity Advanced English) and take some examinations abroad thanks to the opportunities offered by the Erasmus Programme.

Learning skills. The ability to learn is essential because future IT engineers are required to continuously refresh their knowledge, which is crucial in light of the fast development of both concepts and applications that are distinctive of IT Engineering. In addition, the ability to learn may be requested to undertake further studies with a high level of independence. Such skills are assessed: within each single subject, in which students are asked to autonomously explore their knowledge in specific fields of interest; in the drawing up of the paper for the final examination; and possibly within any internship or project activity carried out.

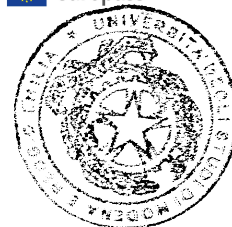


4.3 Programme details, individual credits gained and grades/marks obtained

Educational activities	CFU/ECTS Credits	Grade	Date	Recognition
Additional requirements	0	Passed	19/10/2020	OFA
Advanced English	3	Passed	24/11/2020	A
English	3	Passed	24/11/2020	A
Geometry	9	30 cum laude	19/01/2021	
Fundamentals of Computer Science I and Lab.	9	30 cum laude	27/01/2021	
Calculus I	9	30 cum laude	02/02/2021	
Introduction to Business Management	6	28	08/06/2021	
Applied Mathematics and Statistics	6	30	05/07/2021	
Fundamentals of Computer Science II and Lab.	9	28	08/07/2021	
General Physics	9	26	12/07/2021	
Calculus II	9	30 cum laude	12/01/2022	
Computer Architecture and Lab.	9	30 cum laude	26/01/2022	
Digital Electronics	9	30 cum laude	21/02/2022	
Operating Systems and Lab.	9	29	08/06/2022	
Applied Physics for Informatics	6	28	13/07/2022	
Object-oriented Programming	9	30	20/07/2022	
Databases and Lab.	9	30 cum laude	29/07/2022	
Operative research	6	30 cum laude	16/02/2023	
Computer Languages and Representations (Erasmus)	6	30 cum laude	31/03/2023	EPSTU
Cyber-physical Security	6	30	31/03/2023	EPSTU
Fundamentals of Telecommunications	9	30 cum laude	31/03/2023	EPSTU
Software Engineering and Lab.	9	30 cum laude	31/03/2023	EPSTU
Control systems	9	30 cum laude	07/06/2023	
Computer Networks and Lab.	9	30 cum laude	15/06/2023	
Final examination	3	Three Marks	10/07/2023	

Total CFU/ECTS credits: 180

Legend	
CFU	Credits
A	Recognition of credits acquired at other Italian institutions
EPSTU	Recognition of credits awarded during ERASMUS+ for study Programme
OFA	Recognition of additional learning



Legend	
	requirements

4.4 Grading system and, if available, grade distribution table

For I, II and III cycle programmes the Italian grading system is based on a maximum of 30 points with 18/30 as the lowest passing grade. In case of excellence 30 cum laude may be awarded. Failed exams are not indicated in the student's transcript.

ECTS table of course grades			
ISCED CODE 0714 - L-8 - Information technology engineering - 1° Cycle QF-EHEA /6° LEVEL EQF			
Grade	N° Grade	% Grade	% Cumulative
18	979	7.00	100
19	576	4.00	93.00
20	762	6.00	89.00
21	720	5.00	83.00
22	780	6.00	78.00
23	925	7.00	72.00
24	1163	9.00	65.00
25	1079	8.00	56.00
26	1109	8.00	48.00
27	1123	8.00	40.00
28	1209	9.00	32.00
29	645	5.00	23.00
30	1463	10.00	18.00
30 cum laude	1115	8.00	8.00

Academic years considered: from 01/11/ 2019 to 31/10/ 2022

Total marks: 13648

4.5 Overall classification of the qualification

Final mark: 110 cum laude /110 cum laude .

Date: 18/07/2023

For I, II and III cycle programmes the final grade is based on a maximum of 110 points, with 66/110 as the lowest passing grade. In case of excellence, 110 cum laude may be awarded. The final grade is based on the curriculum as well as on the final exam. For III cycle courses, please refer to the university regulations specific to the course of study.



ECTS table of course grades			
ISCED CODE 0714 - L-8 - Information technology engineering - 1° Cycle QF-EHEA /6° LEVEL EQF			
Mark	N° Grade	% Grade	% Cumulative
82	3	1 %	100
83	5	1 %	99.00
84	4	1 %	98.00
85	15	3 %	97.00
86	13	2 %	94.00
87	13	2 %	92.00
88	15	3 %	90.00
89	16	3 %	87.00
90	28	5 %	84.00
91	21	4 %	79.00
92	20	4 %	75.00
93	17	3 %	71.00
94	22	4 %	68.00
95	19	4 %	64.00
96	25	5 %	60.00
97	26	5 %	55.00
98	17	3 %	50.00
99	33	6 %	47.00
100	24	5 %	41.00
101	15	3 %	36.00
102	20	4 %	33.00
103	19	4 %	29.00
104	16	3 %	25.00
105	15	3 %	22.00
106	15	3 %	19.00
107	13	2 %	16.00
108	13	2 %	14.00
109	14	3 %	12.00
110	4	1 %	9.00
110 cum laude	34	8 %	8.00

Academic years considered: from 01/11/ 2019 to 31/10/ 2022

Total marks: 522

5. Information On The Function Of The Qualification

5.1 Access to further study

Not available

5.2 Professional status (if applicable)

IT engineer focused on the design and production of software applications, and on digital data processing. S/he works as analyst-programmer for the development and integration of any type of software application, both proprietary and open source.

IT engineer focused on the design and production of software applications, and on digital data processing. S/he applies the programming knowledge on imperative languages, objects, and scripts. S/he applies the knowledge of architectures, and the operation of computers, operating systems, databases, Internet, as well as the organisational economic bases offering a view on corporate processes. S/he applies the knowledge of software application development in any field by using the principles and methods of software engineering, with the ability to integrate the knowledge of Web technologies, other than those of traditional architectures, production environments and industrial control, and the latest mobile devices. In addition, s/he is able to apply the basic knowledge of digital electronics, telecommunications, and automated controls, when work is applied to industrial realities not limited to the IT sector. S/he can apply the knowledge linked to CAD design and industrial IT.

IT engineer focused on the design and production of software applications, and on digital data processing. IT companies of any size, industrial, manufacturing and service businesses, public



administrations and freelance professional service as consultant. Greater employment opportunities by widening his/her knowledge by undertaking further studies of Master's Degree Programme in Computer Engineering.

6. Additional information

6.1 Additional information

Activities that are useful for assessing the skills acquired

Mobility Program participation Program: - ERASMUS+ STUDY - UNIVERSITY OF EXETER UNITED KINGDOM - from 19/09/2022 to 14/01/2023

Additional credits gained (also abroad)

Formative activities	CFU/ECTS Credits	Grade	Date	Recognition
Additional requirements	0	Passed	19/10/2020	OFA

6.2 Further information sources

www.unimore.it www.miur.it

7. Certification of the supplement

7.1 Date

30/09/2024

7.2 Name and signature

Dr. Giuseppe Milano
Department of Engineering "Enzo Ferrari"

7.3 Capacity

Registrar's Officer

7.4 Official stamp or seal



8. Information on the national higher education system

The Italian University System

The Italian university system is organised in three cycles, according to the Bologna structure: the main academic degrees are the Laurea (1st cycle), the Laurea Magistrale (2nd cycle) and the Dottorato di Ricerca (3rd cycle). The system also offers other study programmes and related qualifications.

First cycle. This cycle consists exclusively of Corsi di Laurea. These degree programmes provide students with an adequate command of general scientific methods and contents as well as with specific professional skills. The general access requirement is the Italian school leaving qualification awarded after completion of 13 years of schooling and passing the relevant State examination; comparable foreign qualifications may also be accepted. Admission to some degree courses may be based on specific course requirements. The studies last 3 years. The Laurea is awarded to students who have gained 180 ECTS credits (called Crediti Formativi Universitari - CFU) and satisfied all curricular requirements, including the production of a final written paper or equivalent final project. The Laurea gives access to the Corsi di Laurea Magistrale as well as to other 2nd cycle study programmes.

Second cycle. The main degree programmes in this cycle are the Corsi di Laurea Magistrale. They provide education at an advanced level for the exercise of highly qualified activities in specific areas. Access is by a Laurea degree or a comparable foreign degree; admission is based on specific course requirements determined by single universities. The studies last 2 years. The Laurea Magistrale degree is awarded to students who have gained 120 ECTS/CFU credits and satisfied all curricular requirements, including the production and public defence of an original dissertation. Some programmes (namely, those in dentistry, medicine, veterinary medicine, pharmacy, architecture, construction engineering/architecture, law, primary education) are defined "single cycle programmes" (Corsi a ciclo unico); for these programmes access is by the Italian school leaving qualification (or a comparable foreign qualification); admission is based on entrance exams. The studies last 5 years (6 years and 360 ECTS/CFU credits in the cases of medicine and dentistry). A Laurea Magistrale degree is awarded to students who have gained 300 ECTS/CFU credits and satisfied all curricular requirements, including the production and public defence of an original dissertation. A Laurea Magistrale degree gives access to Corsi di Dottorato di Ricerca as well as to other 3rd cycle study programmes.

Third cycle. The main degree programmes in this cycle are Corsi di Dottorato di Ricerca (research doctorate programmes); the students/young researchers enrolled in these programmes will acquire methodologies for advanced scientific research, will be trained in new technologies and will work in research laboratories, wherever appropriate. Access is by a Laurea Magistrale degree (or a comparable foreign degree); admission is based on a competitive exam; studies last at least three years and include the completion and public defence of an original research project.

Other programmes.

- Corsi di Specializzazione. These are 3rd cycle programmes intended to provide students with the knowledge and skills required for the practice of highly qualified professions, mainly in medical, clinical and surgical specialities. Admission is by a Laurea Magistrale degree (or by a comparable foreign degree) and is based on a competitive exam; studies may last from 2 (120 ECTS/CFU credits) to 6 years (360 ECTS/CFU credits) depending on the discipline. The final degree awarded is a Diploma di Specializzazione.

- Corsi di Master Universitario di primo livello. These are 2nd cycle programmes intended to provide students with further specialization or higher continuing education after completion of the



first cycle. Access is by a Laurea degree (or a comparable foreign degree); admission may be subject to additional requirements. Studies last at least 1 year (60 ECTS/CFU credits). The qualification awarded (Master Universitario di primo livello) does not give access to Corsi di Dottorato di Ricerca or to any other 3rd cycle programme since this type of course does not belong to the general requirements established at national level, but it is offered under the autonomous responsibility of each university.

- Corsi di Master Universitario di secondo livello These are 3rd cycle programmes intended to provide students with further specialization or higher continuing education studies after completion of the second cycle. Access is by a Laurea Magistrale degree (or a comparable foreign degree); admission may be subject to additional requirements. Studies last at least 1 year (60 ECTS/CFU credits). The qualification awarded (Master Universitario di secondo livello) does not give access to Corsi di Dottorato di Ricerca or to any other 3rd cycle programmes, since this type of course does not belong to the general requirements established at national level, but it is offered under the autonomous responsibility of each university.

Credits: degree courses are structured in credits (Crediti Formativi Universitari -CFU). University credits are based on the workload students need in order to achieve the expected learning outcomes. Each credit corresponds to 25 hours of student workload, including independent study. The average workload of a full time student is conventionally fixed at 60 credits per year. Thus, the CFU fully coincide with ECTS credits.

Classes of Degree Courses: all degree programmes of Laurea and Laurea Magistrale sharing general educational objectives are grouped into "classes". In developing the specific learning outcomes of single programmes, Universities have to comply with some national requirements for each class concerning the types (and corresponding amount of credits) of teaching learning activities to be included. Degrees belonging to the same class have the same legal value.

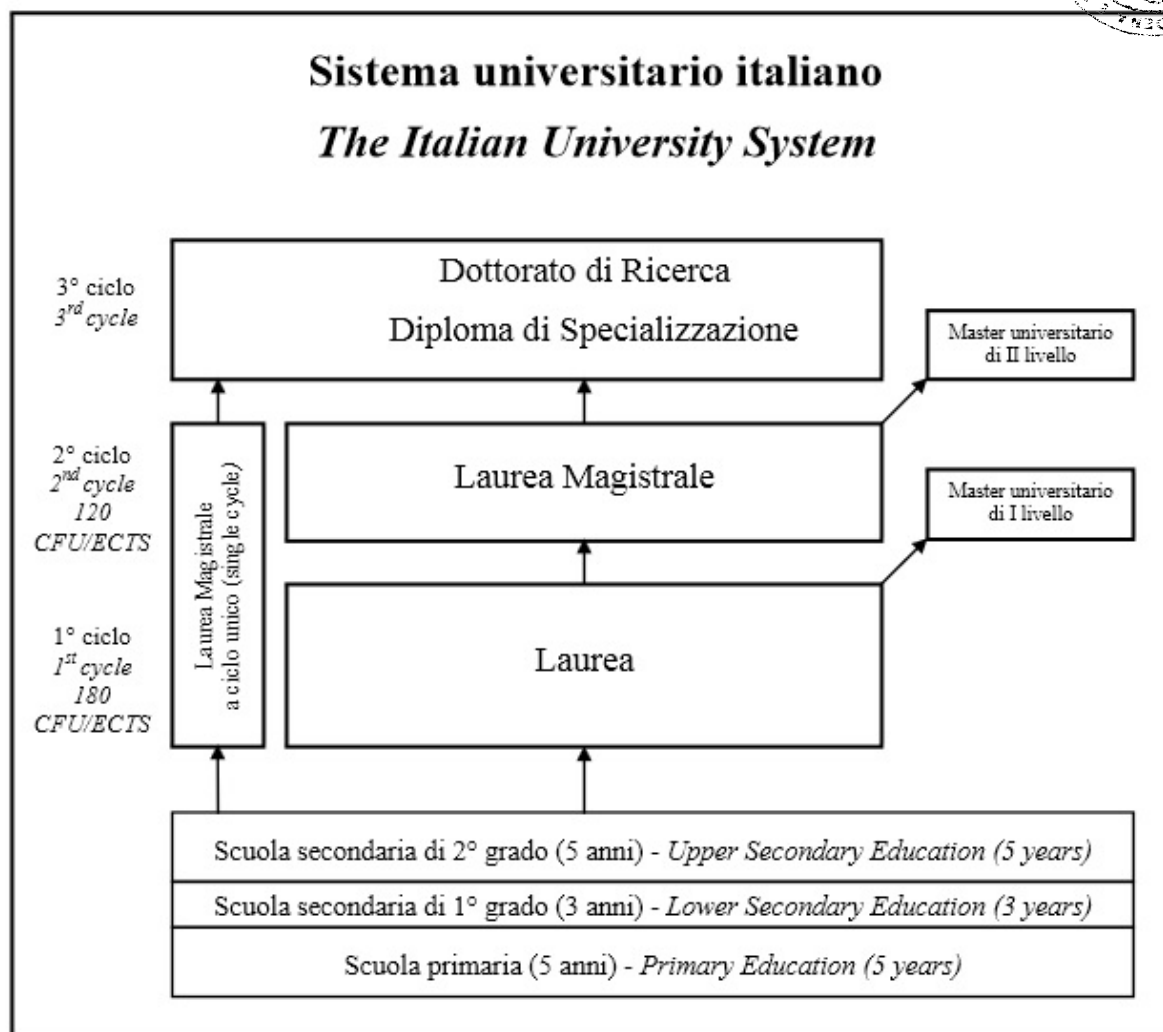
Academic Titles: Those who receive the Laurea are entitled to be called "Dottore", the holders of a Laurea Magistrale have a right to the title of "Dottore Magistrale", the Dottorato di ricerca confers the title of "Dottore di Ricerca" or "PhD".

Joint Degrees: Italian universities are allowed to establish degree programmes in cooperation with Italian and foreign partner universities, on completion of which joint or double/multiple degrees can be awarded.

Further information: Italian Qualifications Framework

Quadro dei Titoli Italiani - QTI

<http://www.quadrodeititoli.it>





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